Rafael Couto Martins

Curriculum Vitae

Laboratoire d'Océanographie Physique et Spatiale (LOPS) - IUEM/UBO/Ifremer.
Technopôle Brest-Iroise, Rue Dumont d'Urville, 29280 Plouzané – France
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Higher Education

2026 – Present Ph.D., Physical Oceanography – Laboratoire d'Océanographie Physique et Spatiale (LOPS) - UBO, Brest.

2024 – 2026 M.Sc., Physical Oceanography – Oceanographic Institute, University of São Paulo (IO USP)

2019 – 2024 B.Sc., Oceanography – Oceanographic Institute, University of São Paulo (IO USP)

Work/Research Experience

2026 - Present PhD student at Laboratoire d'Océanographie Physique et Spatiale (LOPS) - UBO, Brest. Supervisor: Jonathan Gula and Clément Vic

Title: Modelling of Abyssal Currents Over the Mid Atlantic Ridge

Grant by Institut français de recherche pour l'exploitation de la mer (Ifremer)

2025 Master's intern at Laboratoire d'Océanographie Physique et Spatiale (LOPS) - UBO, Brest. Supervisor: Jonathan Gula and Xavier Carton.

Title: The Role of Horizontal Shear in the Evolution of Submesoscale Mixed-Layer Instabilities: An Exploration of the Parameter Space.

International intership project under the grant of FFoundation for Research Support of the State of São Paulo (FAPESP).

2024 - 2026 Master's student at Geophysical Fluid Dynamics and Applications Laboratory at USP. Supervisor: César Barbedo Rocha.

Title: The Role of Lateral Shear and Frontal Curvature in the Onset and Growth of Submesoscale Instabilities.

Master's scholarship under the grant of Foundation for Research Support of the State of São Paulo (FAPESP).

2023 Summer Internship at Exploring the Upper Ocean – Tandon Laboratory, UMass Dartmouth. Supervisor: Amit Tandon and Christian Buckingham.

Title: Classic Wind-driven Models Applied to the South Atlantic Subtropical Gyre.

International internship project under the grant of Foundation for Research Support of the State of São Paulo (FAPESP).

2022 – 2023 Undergrad Research Assistant at Laboratory of Ocean Dynamics (LaDO) – IO USP. Supervisor: Ilson Carlos Almeida da Silveira.

Title: Structure, Seasonal Variability, and Dynamics of the Brazil Current Recirculation Cell.

Scientific Initiation under the grant of Foundation for Research Support of the State of São Paulo (FAPESP).

2021 – 2022 Undergrad Research Assistant at Laboratory of Ocean Dynamics (LaDO) – IO USP. Supervisor: Ilson Carlos Almeida da Silveira.

Title: Sverdrupian Recirculation of the Brazil Current: Structure and Variability on Climatic Scale

Scientific Initiation associated with a project between PETROBRAS and the Oceanography Institute of the University of São Paulo with the administration of FFoundation for the Support of the University of São Paulo (FUSP).

2020 – 2021 Undergrad Research Assistant at Laboratory of Ocean Dynamics (LaDO) – IO USP. Supervisor: Ilson Carlos Almeida da Silveira.

Title: The Brazil Current in a Climate Change Scenario (1980–2016)

Scientific Initiation under the grant of "Unified Scholarship Program (PUB)".

2020 – 2021 Intern at UNESCO Chair on Ocean Sustainability.

2020 Intern at Laboratory of Marine Management, Ecology, and Conservation.

Title: Heuristic and Dialectical Approach to Critical and Interdisciplinary Formation within the Public Policies Program at IOUSP. Supervisor: Alexander Turra.

Culture and Extension Program under the grant of "Unified Scholarship Program (PUB)"

Fellowships

2025 Master's international internship awarded by Foundation for Research Support of the State of São Paulo (FAPESP), Brazil.

2023 Undergraduate's international internship awarded by Foundation for Research Support of the State of São Paulo (FAPESP), Brazil.

2022 - 2023 Undergraduate scholarship awarded by Foundation for Research Support of the State of São Paulo (FAPESP), Brazil.

Conference and Workshops

2026 Ocean Sciences Meeting 2026, Glasgow, Scotland. Poster presentation: The role of lateral shear and frontal curvature in the development of submesoscale instabilities in the upper ocean.

2025 16th Symposium on Waves, Tides, Ocean Engineering and Satellite Oceanography (OMARSAT), Cabo Frio, Rio de Janeiro, Brazil. Poster presentation: Generation and evolution of submesoscale instabilities in the upper ocean.

2024 Ocean Hack Week 2024 at Bigelow Laboratory for Ocean Sciences (Maine - USA).

2024 Empowering the Next Wave of Indian Ocean Prediction: IOPredict-2024 (Hyderabad - India).

2023 15th Symposium on Waves, Tides, Ocean Engineering and Satellite Oceanography (OMARSAT), Cabo Frio, Rio de Janeiro, Brazil. Poster presentation: Structure, Seasonal Variability, and Dynamics of the Brazil Current Recirculation Cell..

Publications

2022 da Silveira, I. C. A., Lazaneo, C. Z., Amorim, J. P. M., Silva. B. M., Bernardo, P. S., **Martins C. R.**, Dottori, M., Belo, W. C., Martins, R. P., Moreira, D. L. Oceanographic conditions of the continental slope and deep waters in Santos Basin: the SANSED cruise (winter 2019). **Ocean and Coastal Research**, OCR-2022-0066.

Teaching Assistant

2022 Ocean Dynamics II, core course of Oceanography (undergraduate), Oceanographic Institute of São Paulo University. August 2022 - December 2022. Professor: Ilson Carlos Almeida da Silveira. Responsibilities: Prepared and corrected short tests to be applied during classes, gave tutoring sessions, and assisted in classroom work.

2023 Synoptic Meteorology and Applications to Oceanography, elective course of Oceanography (undergraduate), Institute of Astronomy, Geophysics and Atmospheric Sciences of the University of São Paulo. August 2023 - December 2023. Professor: Ricardo Camargo. Responsibilities: Prepared and corrected assignments for students, gave tutoring sessions, and assisted in classroom work.

Oceanographic Cruises

2026 Santos Bifurcation (SanBif) 2026 campaign - carried out aboard the adapted research vessel Corrego de Ouro (Macae Navegações), 11 days at sea. Duties: management of ADCP and MicroCAT deployments alongside Sediment Profile Imagery (SPI) equipment, focusing on measurements of bottom temperature and salinity. Assisted with multi-core sediment sampling.

2022 South Atlantic Meridional Overturning Circulation Basin-Wide Array (SAMBA) III – carried out aboard the research vessel Alpha Crucis (University of São Paulo), 17 days at sea. Duties: Assisted in CTD rosette deployments and water sampling; performed salinity analysis using an autosal salinometer; processed underway thermosalinograph (TSG) data to monitor temperature and salinity along the cruise track.

Skills

Languages Portuguese (native)

English (C1 IELTS)

Japanese (N5 JLPT)

Software Python, Julia, MATLAB, LaTeX, Excel, PAST